AP BIOLOGY TOPIC 1.7

Proteins: Amino Acids, Peptide Bonds & 1°–4° Folding

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

SYI-1.B / IST-1.A | R-GROUP CHEMISTRY | 1°–4° HIERARCHY

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.7 (PROTEINS: AMINO ACIDS, PEPTIDE BONDS & 1°–4° FOLDING)

CED Core Vocabulary

Monomers = Amino Acids (20 standard types) — Every amino acid shares an identical backbone bonded to a central asymmetric alpha-carbon (α-C) : Amino Group ($–NH_2$ / $–NH_3^+$): Basic N-terminus end.

Nonpolar (Hydrophobic) R-Groups — Rich in $–CH_3$ / $–CH_2–$ hydrocarbons → avoid water and cluster on the interior core of cytosolic proteins (or face lipid tails in transmembrane domains).

Core Dogma of Protein Biology — DNA Base Sequence → 1° Amino Acid Sequence → 3D Folding (2°/3°/4°) → Specific Function

Triggers — Extreme high temperature (thermal agitation breaks H-bonds/hydrophobic interactions), extreme pH shifts (H^+ / $OH^–$ alters R-group charges, disrupting ionic salt...

Peptide Bonds (Strong Covalent Linkage) — Formed via dehydration synthesis at the ribosome: the carboxyl group (–COOH) of one amino acid reacts with the amino group ($–NH_2$) of the next amino acid,...

Polar Uncharged (Hydrophilic) R-Groups — Contain –OH , $–NH_2$, or –SH → form hydrogen bonds with water or other polar R-groups on the protein exterior.

Major Protein Functions — Enzymatic catalysis (lowers E_a), structural support (keratin, collagen), membrane transport (channels/pumps), cell signaling (receptors, insulin), defense...

What Breaks vs. What Survives — Denaturation disrupts weak 2°, 3°, and 4° interactions (unfolding the active site), while 1° covalent peptide bonds remain intact ! Sometimes reversible...

§2. AMINO ACIDS & PEPTIDE BOND FORMATION

N-Terminus → C-Terminus

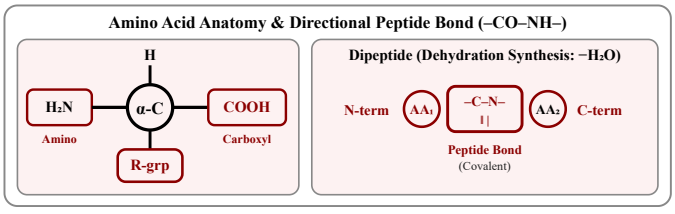
HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Amino Acids & Peptide Bond Formation ➔

Monomers = Amino Acids (20 standard types) ➔

Peptide Bonds (Strong Covalent Linkage)

- **Monomers = Amino Acids (20 standard types):** Every amino acid shares an identical backbone bonded to a central asymmetric **alpha-carbon (** **α-C** **)**:
  - **Amino Group (** **$–NH_2$** / **$–NH_3^+$** **):** Basic N-terminus end.
  - **Carboxyl Group (** **–COOH** / **$–COO^–$** **):** Acidic C-terminus end.
  - **Hydrogen Atom (** **–H** **)** + a variable **R Group (Side Chain)**.
- **Peptide Bonds (Strong Covalent Linkage):**
  - Formed via **dehydration synthesis** at the ribosome: the **carboxyl group (** **–COOH** **)** of one amino acid reacts with the **amino group (** **$–NH_2$** **)** of the next amino acid, releasing **1** **H_2O** .
  - Creates a directional **polypeptide chain** synthesized from **N-terminus (** **$–NH_2$** **)** → **C-terminus (** **–COOH** **)**.



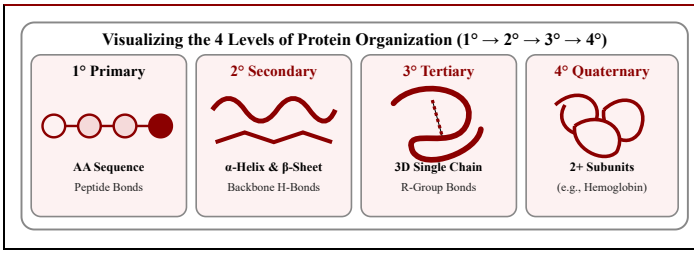
[CORE AXIOM / REMEMBER] **Peptide bond = "link between beads"**; always write or state: **"a covalent peptide bond forms between the carboxyl group of one amino acid and the amino group of the next via a dehydration reaction."**

[COMMON AP EXAM FRQ TRAP] **Confusing peptide bonds with hydrogen bonds: Peptide bonds** are **strong covalent bonds** that hold the primary backbone together (NOT broken by heat/pH denaturation!).

§4. THE 4 LEVELS OF PROTEIN STRUCTURE (1° → 4°)

Hierarchy of Folding

LEVEL	STRUCTURAL DEFINITION	STABILIZING BONDS / INTERACTIONS
Primary (1°)	Unique **linear sequence of amino acids** (N-terminus → C-terminus), encoded by mRNA codons.	**Covalent peptide bonds (** **–CO–NH–** **)** between backbone monomers.
Secondary (2°)	Local regular coils & folds: **α-helix** (spiral) and **β-pleated sheet** (parallel/antiparallel ribbons).	**Hydrogen bonds along the polypeptide backbone (** **C=O** oxygen to **N–H** hydrogen; *NOT R-groups!*).
Tertiary (3°)	Overall **3D folded conformation** of a single polypeptide chain (creates enzyme active sites).	**R-group interactions:** Hydrophobic clustering, H-bonds, ionic salt bridges, & covalent **disulfide bridges (** **–S–S–** **)**.
Quaternary (4°)	Assembly of **≥2 polypeptide subunits** into one functional multimeric protein complex.	Same noncovalent & disulfide interactions as 3°, acting **between separate subunits**.



[EXAMPLE & BIOLOGICAL APPLICATION] • **Sickle-Cell Anemia (1° → 4° Cascade):** A single amino acid substitution at position 6 of β-globin replaces hydrophilic **Glutamic Acid (charged)** with hydrophobic **Valine (nonpolar)** → abnormal hydrophobic patch causes hemoglobin tetramers (4°) to aggregate into rigid fibers that sickle red blood cells.

• **3° & 4° Examples: Enzyme active sites** depend on precise 3° folding; **Hemoglobin** (2 α + 2 β subunits) & **Collagen** (3 helical chains) exhibit 4° structure.

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§3. R GROUPS (SIDE CHAINS) DETERMINE FOLDING BEHAVIOR

Chemical Personality

- While the N–C–C backbone is identical in all amino acids, the **variable R group** dictates chemical properties and 3D folding:
  - **Nonpolar (Hydrophobic) R-Groups:** Rich in **$-CH_3$ / $-CH_2-$** hydrocarbons → avoid water and **cluster on the interior core** of cytosolic proteins (or face lipid tails in transmembrane domains).
  - **Polar Uncharged (Hydrophilic) R-Groups:** Contain **$-OH$**, **$-NH_2$**, or **$-SH$** → form **hydrogen bonds** with water or other polar R-groups on the protein exterior.
  - **Charged (Ionic: Acidic (**$-COO^-$**) or Basic (**$-NH_3^+$**) R-Groups:** Hydrophilic; attract opposite charges to form **ionic bonds (salt bridges)**.
  - **Sulfhydryl (**$-SH$** in Cysteine):** Two Cysteine side chains oxidize to form a **covalent disulfide bridge (**$-S-S-$**)**.

[CORE AXIOM / REMEMBER] **R groups = "personality of amino acids"** (dictates whether that region folds inward away from water or outward toward water).

[AP EXAM FRQ SCORING STRATEGY] **Mutation FRQ Strategy:** When an AP question asks how a missense mutation affects a protein, compare the **R-group chemistry**! Replacing a *hydrophobic* amino acid with another *hydrophobic* one may be silent/neutral, whereas replacing a *hydrophilic/charged* R-group with a *nonpolar* R-group alters 3° folding and destroys active-site function!

§5. STRUCTURE–FUNCTION & PROTEIN DENATURATION

Active Site Loss

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Structure–Function & Protein Denaturation ➔ **Core Dogma of Protein Biology** ➔ **Major Protein Functions**

- **Core Dogma of Protein Biology:**

DNA Base Sequence → **1° Amino Acid Sequence** → **3D Folding (2°/3°/4°)** → **Specific Function**

- **Major Protein Functions:** **Enzymatic catalysis** (lowers **E_a**), **structural support** (keratin, collagen), **membrane transport** (channels/pumps), **cell signaling** (receptors, insulin), **defense** (antibodies), and **motility** (actin/myosin).
- **Denaturation (Loss of Native 3D Shape & Function):**
  - **Triggers:** Extreme **high temperature** (thermal agitation breaks H-bonds/hydrophobic interactions), extreme **pH shifts** (**H^+ / OH^-** alters R-group charges, disrupting ionic salt bridges), or high **salinity/urea**.
  - **What Breaks vs. What Survives:** Denaturation disrupts weak **2°, 3°, and 4° interactions** (unfolding the active site), while **1° covalent peptide bonds remain intact**! Sometimes reversible (renaturation) if gentle, or permanent (e.g., boiling an egg).

[AP EXAM FRQ SCORING STRATEGY] **The 3-Link FRQ Chain:** Whenever explaining mutations or environmental shifts in proteins, you MUST link all three: **1) Primary sequence / R-group interactions → 2) 3D folding / active site shape → 3) Biological function / catalytic activity**!

[MECHANISTIC NOTE] **AP CED Scope Note:** Memorizing the exact chemical structures of the 20 individual amino acid R-groups is NOT required on the AP Exam — you will be given structures or properties and asked to classify them as **nonpolar, polar, or charged/ionic**!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.7 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 1.7: Proteins: Amino Acids, Peptide Bonds & 1°–4° Folding ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, (n − 1) covalent bonds form and (n − 1) H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.7 to prove why the claim holds.

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